

C2_06 Gate C Protected Review - Astra RC4

Clinic Appointments. Final fresh reassessment; open only after a saved genuine Gate C attempt.

1. Objective fixture controls

Control	Expected	Why it matters
Processes	95 appointments / 63 billing rows	Appointment and billing are separate facts/event dates even when linkage is one-to-one for completed cases.
Status	63 Completed / 32 Cancelled	Population definition matters before billing/completion analysis.
Billing	63 unique appointments; 0 billing orphans; total 76,728.69	Clean linkage does not remove the need for provider history and target grain.
Unknown provider	P99 on 1 Completed appointment	A completed business event has unresolved dimension attribution.
History	P04 General -> Specialist effective 2026-04-20	Point-in-time specialty needs Type 2 versioning.
P04 evidence	7 appointments before / 7 after; 5 completed in each side	Current-only specialty would rewrite historical reporting.
Target	12 unique Specialty-Month rows; total 1,242	Target stays separate at specialty-month grain.
Date roles	Appointments Mar/Apr/May/Jun 30/30/23/12; billing 19/20/15/9	Appointment and billing period questions must name the date role.

2. Expected reasoning path

Model appointment and billing processes at their own grains; use compatible dimensions to compare them.

Resolve P04 specialty history with non-overlapping effective versions and fact-date lookup.

Keep Specialty-Month targets separate; P99 must remain as unknown/exception until governed resolution.

Trace source -> history mapping -> facts/targets -> KPI, and block release on unresolved critical grain/history/FK controls.

3. Critical failures

- Billing rows are treated as all appointments.
- P04 Specialist value overwrites pre-20-Apr history.
- Specialty target repeats per appointment/billing row.
- P99 Completed appointment disappears.
- AppointmentDate and BillingDate are conflated.
- Quality check is documented but not release-blocking.

4. Competency floors

Competency	Required proof
Process / grain	Facts, dimensions and targets represent declared processes/grains; no fan-out or repeated target.
Keys / SCD history	Business/surrogate keys, unknown-member handling and non-overlapping historical intervals are correct.
Layer / store / latency	ETL/ELT, replay layers, platform and batch/stream choice follow requirements and recovery needs.
Lineage / metadata	Source-to-KPI path, owner/grain/schema/rule metadata are sufficient to explain the output.
Quality / release	Reconciliations and integrity checks have severity/owner/action; blocking failure stops publication.

Changed repair proof

This protected review opens only after a saved genuine Gate C attempt. Close it, complete a changed fresh retry on new provider/history/target/date-role evidence, and save it. If instability remains, repair only the named weak competency; do not recycle prior Gate designs.